

Affinoid spaces

Fix a non-archimedean completed valued field $(\mathbf{k}, |\cdot|)$ throughout this section. Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. It should be viewed as the non-archimedean analogue of an affine variety in algebraic geometry or an affine scheme in scheme theory. The goal of this section is to define a suitable Grothendieck topology on X such that we can define a structure sheaf of analytic functions on X , and then study its basic properties.

1 Affinoid domains

On a general open subset $U \subseteq X$, the ring of analytic functions on U may behave badly.

Yang: To be continued.

Definition 1. A closed subset $V \subseteq X$ is called an *affinoid domain* if there exists a $\mathbf{k}$ -affinoid algebra A_V and a morphism of $\mathbf{k}$ -affinoid algebras $\theta : A \rightarrow A_V$ satisfying the following universal property: for every bounded homomorphism of $\mathbf{k}$ -affinoid algebras $\psi : A \rightarrow B$ such that the induced map on spectra $\mathcal{M}(\psi) : \mathcal{M}(B) \rightarrow X$ has its image contained in V , there exists a unique bounded homomorphism $\varphi : A_V \rightarrow B$ such that the following right diagram commutes:

$$\begin{array}{ccc} & V & \\ \swarrow & & \nwarrow \\ X & \xleftarrow{\mathcal{M}(\psi)} & \mathcal{M}(B) \end{array} \quad \rightsquigarrow \quad \begin{array}{ccc} & A_V & \\ \theta \nearrow & & \searrow \varphi \\ A & \xrightarrow{\psi} & B \end{array}$$

In this case, we say that V is represented by the $\mathbf{k}$ -affinoid algebra A_V . If A_V is strict, then we say that V is a strict affinoid domain.

Slogan A closed subset $V \subset X$ is an affinoid domain if the functor “ $\text{Mor}(-, V)$ ” is representable by a $\mathbf{k}$ -affinoid algebra A_V .

There are several important classes of affinoid domains.

Construction 2. Let $f = (f_1, \dots, f_n), g = (g_1, \dots, g_m)$ be two tuples of elements in A and $r = (r_1, \dots, r_n), s = (s_1, \dots, s_m)$ be two tuples of positive real numbers. Consider the following closed subset of X :

$$X(\underline{f/r}; \underline{s/g}) := \{x \in X : |f_i(x)| \leq r_i, |g_j(x)| \geq s_j, 1 \leq i \leq n, 1 \leq j \leq m\}.$$

Such a closed subset is called a *Laurent domain* of X . If $m = 0$, then it is called a *Weierstrass domain* of X , denoted by

$$X(\underline{f/r}) := \{x \in X : |f_i(x)| \leq r_i, 1 \leq i \leq n\}.$$

More generally, let $f = (f_1, \dots, f_n), g$ be elements in A such that the ideal generated by them is the whole algebra A . Set $p = (p_1, \dots, p_n)$ be a tuple of positive real numbers. We define the following closed subset of X :

$$X(\underline{(f/g)/p}) := \{x \in X : |f_i(x)| \leq p_i |g(x)|, 1 \leq i \leq n\}.$$

| Such a closed subset is called a *rational domain* of X .

Proposition 3. Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. Then Weierstrass domains, Laurent domains, and rational domains of X are affinoid domains. They are represented by $A\{f/r\}$, $A\{f/r; g/s\}$, and $A\{(f/g)/p\}$ in Yang: ref, respectively.

| *Proof.* Yang: To be added. □

We have a sequence of inclusion:

$$\{\text{Weierstrass domains}\} \subseteq \{\text{Laurent domains}\} \subseteq \{\text{Rational domains}\} \subseteq \{\text{Affinoid domains}\}.$$

Yang: Why Laurent domain is rational domain?

Yang: Is these domain open? Yang: In the sence of Tate, they are.

Proposition 4. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . Then the natural map on spectra $\mathcal{M}(\theta) : \mathcal{M}(A_V) \rightarrow X$ induced by the representing homomorphism $\theta : A \rightarrow A_V$ has its image equal to V , and induces a homeomorphism $\mathcal{M}(A_V) \cong V$.

| *Proof.* Yang: To be added. □

Proposition 5. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . Then the natural bounded homomorphism $\varphi : A \rightarrow A_V$ is flat.

| *Proof.* Yang: To be added. □

Proposition 6. Let $f : Y = \mathcal{M}(B) \rightarrow X = \mathcal{M}(A)$ be a morphism of $\mathbf{k}$ -affinoid spaces induced by a bounded homomorphism of $\mathbf{k}$ -affinoid algebras $\phi : A \rightarrow B$. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . Then the inverse image $f^{-1}(V) \subseteq Y$ is an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra $B \hat{\otimes}_A A_V$. Yang: To be revised.

| *Proof.* Yang: To be added. □

Proposition 7. Let $V_1, V_2 \subseteq X$ be two affinoid domains represented by the $\mathbf{k}$ -affinoid algebras A_{V_1}, A_{V_2} , respectively. Then the intersection $V_1 \cap V_2 \subseteq X$ is an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra $A_{V_1} \hat{\otimes}_A A_{V_2}$. Yang: To be checked.

| *Proof.* Yang: To be added. □

Proposition 8. Let $x \in X$ be a point. Then the collection of strict affinoid domains of X containing x forms a fundamental system of neighborhoods of x . Explicitly, for any open neighborhood $U \subseteq X$ of x , there exists a strict affinoid domain $V \subseteq X$ such that $x \in V \subseteq U$. Yang: To be checked.

Theorem 9 (Gerritzen-Grauert theorem). Let $V \subseteq X$ be an affinoid domain. Then there exist finitely many rational domains $V_1, \dots, V_m \subseteq X$ such that $V = V_1 \cup \dots \cup V_m$. Yang: To be checked.

| *Proof.* Yang: To be added. □

Definition 10. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . Suppose that $A_V \hat{\otimes} \mathbf{K}$ is strictly $\mathbf{K}$ -affinoid for some completed valued field extension $\mathbf{K}/\mathbf{k}$. The *dimension* of

V is defined as the Krull dimension of the affinoid algebra $A_V \hat{\otimes} \mathbf{K}$:

$$\dim(V) := \dim_{\text{Krull}}(A_V \hat{\otimes} \mathbf{K}).$$

By [Proposition 28](#), the definition is independent of the choice of $\mathbf{K}$.

2 The Grothendieck topology and structure sheaf

Theorem 11 (Tate's Acyclicity Theorem). Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. Let $\{V_i\}_{i \in I}$ be a finite covering of X by affinoid domains. For each finite subset $J \subseteq I$, set $V_J := \bigcap_{j \in J} V_j$, which is an affinoid domain by [Proposition 7](#). Then the augmented Čech complex

$$0 \rightarrow A \rightarrow \prod_{i \in I} A_{V_i} \rightarrow \prod_{\substack{J \subseteq I \\ |J|=2}} A_{V_J} \rightarrow \prod_{\substack{J \subseteq I \\ |J|=3}} A_{V_J} \rightarrow \dots$$

is exact. [Yang: To be checked.](#)

Proof. [Yang: To be added.](#) □

Definition 12. A *special domain* of X is a finite union of affinoid domains of X . Let $V \subseteq X$ be a special domain, say $V = V_1 \cup \dots \cup V_m$ with each V_i an affinoid domain. We define

$$A_V := \ker \left(\prod_{i=1}^m A_{V_i} \rightarrow \prod_{1 \leq i < j \leq m} A_{V_i \cap V_j} \right).$$

[Yang: To be checked.](#)

Proposition 13. Let $V \subseteq X$ be a special domain, say $V = V_1 \cup \dots \cup V_m$ with each V_i an affinoid domain. Then V is an affinoid domain if and only if the $\mathbf{k}$ -algebra A_V defined in [Definition 12](#) is a $\mathbf{k}$ -affinoid algebra. In this case, V is represented by the $\mathbf{k}$ -affinoid algebra A_V . [Yang: To be added.](#)

Definition 14. The *site of special domains* on X , is defined as follows:

- the underlying category has as objects the special domains $V \subseteq X$, and the morphisms are given by the inclusion maps;
- a covering of an objects $V \subseteq X$ is a finite family of affinoid domains $\{V_i\}_{i \in I}$ such that $V = \bigcup_{i \in I} V_i$.

The *structure sheaf on site of special domains* is given by assigning to each special domain $V \subseteq X$ the ring A_V . [Yang: To be revised.](#)

Definition 15. For every open subset $U \subseteq X$, we define the ring of analytic functions on U as follows:

$$\mathcal{O}_X(U) := \varprojlim_{V \subseteq U} A_V \in \mathbf{Alg}_{\mathbf{k}},$$

where the limit is taken over all special domains $V \subseteq U$, and A_V is the $\mathbf{k}$ -algebra representing the special domain V . [Yang: To be revised.](#)

Remark 16. Note that in Definition 14 we have defined a site on X consisting of special domains, and in Definition 15 we have defined a structure sheaf on the usual topological space X . These two sheaves are closely related, and also has some differences.

Yang: on special site, the assigning algebra is a normed algebra, $\mathbf{k}$ -affinoid algebra. On open subset, the assigning algebra is even not necessarily normed.

Yang: If a special subset $V \subseteq X$ is open, then they coincide?

Yang: On a point, they are the same? Yang: Yes, this is right.

Yang: To be revised.

Definition 17. The morphism of $\mathbf{k}$ -affinoid spaces is defined as the morphism $(f, f^\#)$ of locally ringed spaces satisfying that

- (a) the underlying map $f : Y \rightarrow X$ is continuous with respect to the spectral topology;
- (b) the morphism $f^\# : \mathcal{O}_X \rightarrow f_* \mathcal{O}_Y$ is bounded, that is, for every special domains $V \subseteq X$ and $W \subseteq Y$ such that $f(W) \subseteq V$, the induced homomorphism $A_V \rightarrow A_W$ is bounded.

We denote by Yang: the category of $\mathbf{k}$ -affinoid spaces with morphisms defined as above.

Proposition 18. The category of $\mathbf{k}$ -affinoid spaces is equivalent to the category of $\mathbf{k}$ -affinoid algebras with bounded homomorphisms via the functor $\mathcal{M}$.

Let M be an A -module, we can associate to M a sheaf $\mathcal{O}_X(M)$ of $\mathcal{O}_X$ -modules on X by setting

$$\Gamma(U, \mathcal{O}_X(M)) := \varprojlim_{V \subseteq U} A_V \otimes_A M,$$

where the limit is taken over all special domains $V \subseteq U$.

Proposition 19. The functor $M \mapsto \mathcal{O}_X(M)$ induces an equivalence of categories between the category of finite A -modules and the category of coherent $\mathcal{O}_X$ -modules on X .

3 Boundary and interior

Let A, B be two $\mathbf{k}$ -affinoid algebras, and let $f : Y = \mathcal{M}(B) \rightarrow X = \mathcal{M}(A)$ be a morphism of $\mathbf{k}$ -affinoid spaces induced by a bounded homomorphism of $\mathbf{k}$ -affinoid algebras $\phi : A \rightarrow B$.

Definition 20. The *relative interior* of Y over X is defined as

$$(Y/X) := \{y \in Y : \text{there exists an admissible surjective } A \rightarrow B \text{ inducing } y\}.$$

Yang: To be revised.

Definition 21.

4 Some local properties

In this subsection, we will study some local properties of affinoid spaces. As in algebraic geometry, the local ring at a point $x \in X$ is defined as

$$\mathcal{O}_{X,x} := \varinjlim_{x \in U} \mathcal{O}_X(U),$$

where the limit is taken over all open neighborhoods U of x in X .

Proposition 22. We have

$$\mathcal{O}_{X,x} \cong \varinjlim_{\substack{x \in V \\ \text{affinoid}}} A_V \cong \varinjlim_{\substack{x \in V \\ \text{special}}} A_V.$$

Yang:

Theorem 23. The ring $\mathcal{O}_{X,x}$ at any point $x \in X$ is a noetherian local ring. And it is faithfully flat over the local ring $\mathcal{O}_{\text{Spec } A, \wp_x}$.

Proof. Yang: To be added. □

In algebraic geometry, there is an amazing fact we often overlook: the residue field of the local ring at a point x of an affine variety is isomorphic to the residue field of the local ring at x of any open neighborhood of x . This fails in the non-archimedean analytic geometry, see [Example 24](#).

Example 24. Let $A = k\{T\}$ and $X = \mathcal{M}(A)$.

Yang: To be added.

By [Example 24](#), the residue field $\text{Frac } A/\wp_x$ may not be a good invariant of the point x . This seems a good reason to consider the completed residue field $\mathcal{H}(x)$ instead, by [Proposition 25](#).

Proposition 25. Let $x \in X$ be a point, and let V be an affinoid domain containing x represented by the $\mathbf{k}$ -affinoid algebra A_V . Then the completed residue field $\mathcal{H}(x)$ is isomorphic to the completion of the residue field $\text{Frac } A_V/\wp_x$ with respect to the norm induced by the point x .

Furthermore, the completed residue field $\mathcal{H}(x)$ contains the residue field $\mathcal{O}_{X,x}/\mathfrak{m}_x$ as a dense subfield.

Proof. Yang: To be added. □

Definition 26. Let $\mathcal{P}$ be one of the following properties: regular, normal, reduced, Gorenstein, Cohen-Macaulay, complete intersection, etc. We say that the affinoid space X has a local property $\mathcal{P}$ at a point $x \in X$ if the local ring $\mathcal{O}_{X,x}$ has the property $\mathcal{P}$. Yang: To be added.

Using $\mathcal{P}(X)$ to denote the set of points $x \in X$ such that X has the property $\mathcal{P}$ at x , we have the following result.

Theorem 27. Let X be an affinoid space, and let $\mathcal{P}$ be one of the following properties: regular, normal, reduced, Gorenstein, Cohen-Macaulay, complete intersection. Then the locus of points $x \in X$ such that X has the property $\mathcal{P}$ at x is a Zariski open subset of X . More explicitly, we have $\mathcal{P}(X) = \text{Ker}^{-1}(\mathcal{P}(\text{Spec } A))$.

Appendix

Proposition 28. Let A be a strictly $\mathbf{k}$ -affinoid algebra. For any completed valued field extension $\mathbf{K}/\mathbf{k}$, we have

$$\dim_{\mathbf{K}\text{rull}}(A) = \dim_{\mathbf{K}\text{rull}}(A \hat{\otimes}_{\mathbf{k}} \mathbf{K}).$$

Yang: To be checked.

DRAFT